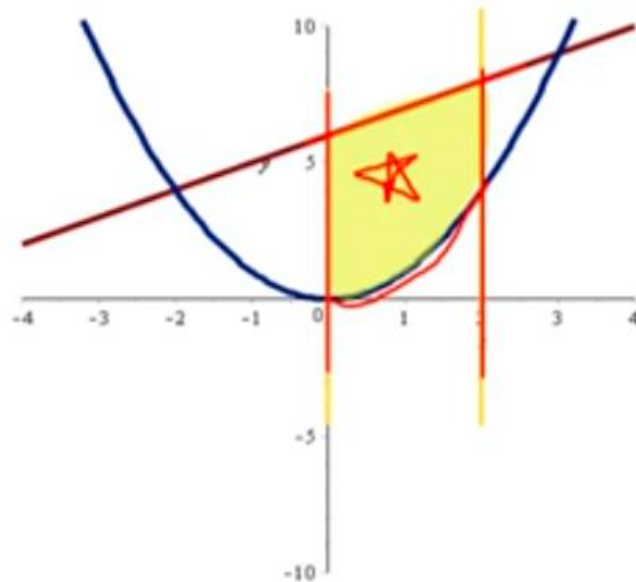


Area Between Two Curves

If f and g are continuous functions on the interval $[a, b]$, and if $f(x) \geq g(x)$ for all x in $[a, b]$, then the area of region bounded above by $y = f(x)$, below by $y = g(x)$, on the left by the line $x = a$, and on the right by the line $x = b$ is

$$A = \int_a^b [f(x) - g(x)] dx.$$

Graph:



Example 1. Find the area of the region bounded above by $f(x) = x + 6$, bounded below by $g(x) = x^2$, and bounded on the sides by the lines $x = 0$ and $x = 2$.

Solution.

$$\begin{aligned} A &= \int_a^b [f(x) - g(x)] dx \\ &= \int_0^2 [(x + 6) - x^2] dx \\ &= \left[\frac{x^2}{2} + 6x - \frac{x^3}{3} \right]_0^2 \\ &= \left[\left(\frac{(2)^2}{2} + 6(2) - \frac{(2)^3}{3} \right) - \left(\frac{(0)^2}{2} + 6(0) - \frac{(0)^3}{3} \right) \right] \\ &= \frac{4}{2} + 12 - \frac{8}{3} = \frac{68}{6} = \frac{34}{3} \end{aligned}$$

Example 2. Find the area of the region enclosed between the curves $f(x) = x + 6$ and $g(x) = x^2$.

Solution.

Step 1. (Limits of Integration)

In order to find limits of integration we find points of intersections of the curves $y = x + 6$ and $y = x^2$:

$$\begin{aligned} x^2 &= x + 6 \\ x^2 - x - 6 &= 0 \\ x^2 - 3x + 2x - 6 &= 0 \\ x(x - 3) + 2(x - 3) &= 0 \\ (x + 2)(x - 3) &= 0 \end{aligned}$$

It follows that the limits of integration are $a = -2$ and $b = 3$.

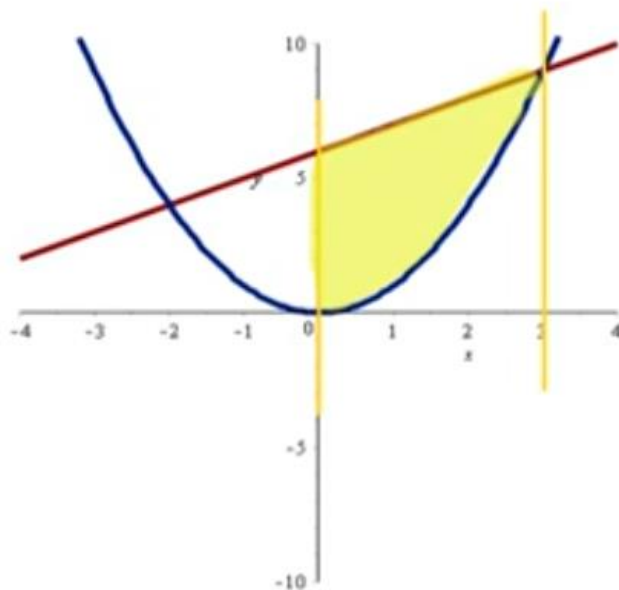
Step 2. (Upper and Lower Curves)

In order to decide which curve lies above the other, we take a test point in the interval $[-2, 3]$ and find values of f and g at that point. The curve with larger value will lie above the other.

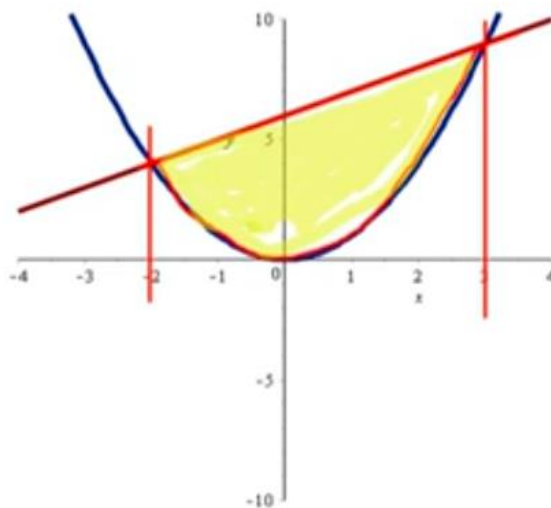
If we take $x = 0$, then $f(0) = 6$ and $g(0) = 0$. Since $f(0) > g(0)$, the graph of f lies above the graph of g .

Step 3. (Area between Curves)

$$\begin{aligned}
 A &= \int_a^b [f(x) - g(x)] dx \\
 &= \int_{-2}^3 [(x+6) - x^2] dx \\
 &= \left[\frac{x^2}{2} + 6x - \frac{x^3}{3} \right]_{-2}^3 \\
 &= \left[\left(\frac{(3)^2}{2} + 6(3) - \frac{(3)^3}{3} \right) - \left(\frac{(-2)^2}{2} + 6(-2) - \frac{(-2)^3}{3} \right) \right] \\
 &= \left[\frac{9}{2} + 18 - \frac{27}{3} \right] - \left[\frac{4}{2} - 12 - \frac{-8}{3} \right] \\
 &= \frac{27 + 108 - 54}{6} - \frac{12 - 72 + 16}{6} \\
 &= \frac{81}{6} - \frac{-44}{6} \\
 &= \frac{125}{6} \quad \bullet
 \end{aligned}$$



$$\begin{aligned}
&= \left[\frac{9}{2} + 18 - \frac{27}{3} \right] - \left[\frac{4}{2} - 12 - \frac{-8}{3} \right] \\
&= \frac{27 + 108 - 54}{6} - \frac{12 - 72 + 16}{6} \\
&= \frac{81}{6} - \frac{-44}{6} \\
&= \frac{125}{6}
\end{aligned}$$



Example 3. Find the area of the region enclosed between the curves $f(x) = 2x$ and $g(x) = x^2$.

Solution.

Step 1. (Limits of Integration)

In order to find limits of integration we find points of intersections of the curves $y = 2x$ and $y = x^2$:

$$\begin{aligned}
x^2 &= 2x \\
x^2 - 2x &= 0 \\
x(x - 2) &= 0
\end{aligned}$$

It follows that the limits of integration are $a = 0$ and $b = 2$.

Step 2. (Upper and Lower Curves)

In order to decide which curve lies above the other, we take a test point in the interval $[0, 2]$ and find values of f and g at that point. The curve with

larger value will lie above the other.

If we take $x = 1$, then $f(1) = 1$ and $g(1) = 2$. Since $g(1) > f(1)$, the graph of g lies above the graph of f .

Step 3. (Area between Curves)

$$\begin{aligned} A &= \int_a^b [g(x) - f(x)] dx \\ &= \int_0^2 [2x - x^2] dx \\ &= \left[x^2 - \frac{x^3}{3} \right]_0^2 \\ &= \left[(2)^2 - \frac{(2)^3}{3} \right] - \left[(0)^2 - \frac{(0)^3}{3} \right] \\ &= 4 - \frac{8}{3} - 0 \\ &= \frac{4}{3} \end{aligned}$$

Graph:

